Forward Feynman Gives Schrödinger, Reversible Feynman Gives QTSI

Executive summary for readers in quantum foundations

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The question

Quantum mechanics supplies no microscopic definition of the direction of time. The dynamics is invariant under antiunitary reversal, the Born rule is not a dynamical statement, and the asymmetry recovered at the level of ensembles is thermodynamic in origin and macroscopic in scale. We take that absence as the point of departure and ask what a sum over histories yields when the temporal orientation of a path is not fixed in advance. The answer divides into two theories, and the division is sharper than expected: they are separated by the phase of a single complex number, and everything else follows.

The result reported is not a new reading of the measurement problem. It is that the postulates ordinarily granted at the start of a quantum mechanics course are theorems of a path sum, in two versions, and that the version obtained by refusing the temporal asymmetry is the two-component time-symmetric formalism.

The forward theorem: three postulates from one geometric assumption

We re-examine the equivalence between Feynman's sum over histories and the Schrödinger equation, and we locate the assumption that both directions of the equivalence require. In the short-time construction the amplitude at $t + \varepsilon$ is built from the amplitude at t by integrating over exactly one intermediate position per path. In the Lie-Trotter product the propagator is cut at intermediate times, and each cut contributes exactly one integration variable. Both steps presume that a contributing history is a graph $x(t)$, single-valued in the time coordinate.

From that presumption three familiar statements follow in sequence rather than independently. Only one earlier slice is required, so the equation of motion is first order in the time derivative. A first-order equation with a self-adjoint generator conserves the integral of $|\psi|^2$. That conserved quantity is a sum of squares, hence positive, hence available for the probability reading that the Born rule requires. Monotonic time, first order, unitarity in the ordinary sense, and the positivity behind the Born rule are four names for one geometric fact about the histories being summed. Removing the fact removes all four together, which is what the second half of the derivation carries out.

Withdrawing the assumption

On a lattice with a reversible update rule, stepping backward is exactly as legitimate an operation as stepping forward, so paths that reverse their sense of traversal of time are admissible. A folded history crosses a given instant three times, or five, or any odd number, and no single crossing point remains to integrate over. The semigroup property

$$K(x'', t''; x, t) = \int dx'\, K(x'', t''; x', t')\, K(x', t'; x, t)$$

fails for $t < t' < t''$. Losing that identity is losing first order in time, and the conclusion follows from the geometry of a folded line with no dynamical input whatever.

Two components are then forced by arithmetic. A folded history carries more than one sense of traversal at a given instant, so the amplitude cannot be recorded as one number per instant. The recursion agrees: amplitudes on cell n − 1 reach cell n + 1, the update becomes a three-term relation, and a three-term relation requires two initial data. The doubling is not an interpretive move.

A negative result that we record rather than conceal

It is tempting to fix the conserved bilinear by counting. Cut a folded worldline with a time slice, count crossings with sign, and observe that inserting a fold pair raises the total count by two while leaving the signed count unchanged. The signed count is an invariant of the worldline and the total count is not, which establishes that the positive bilinear cannot be the density of anything the folding conserves.

The counting does not establish that a fold vertex must preserve $Q = |\psi+|^2 - |\psi-|^2$, and we state plainly why not. A fold vertex leaves the signed count unchanged whatever amplitude it carries, since folds enter in pairs and a pair contributes zero. A quantity conserved for every possible vertex weight cannot select among vertex weights. There is also a difference of type: the signed count is an integer attached to a geometric object and belongs to the specification of which configuration is being summed, while Q is a real quadratic form on a two-dimensional complex amplitude space built from the sums themselves. No canonical map carries the first into the second. An independent argument is required, and the counting retains only the role it can support, which is to say what the conserved quantity means once it has been derived.

The central theorem: reversibility fixes the vertex

The vertex is fixed by reversibility stated at the level of amplitudes rather than at the level of probabilities. The operation reversing the sense of traversal exchanges the two branches and conjugates amplitudes, since it reverses the sign of the phase accumulated along a segment. It is the antiunitary

$$\Theta = \sigma 1\, K, \quad \Theta\, M\, \Theta^{-1} = \sigma 1\, M^{*}\, \sigma 1$$

with K complex conjugation. A rule that does not distinguish the two senses of traversal commutes with Θ. The propagation step does so identically and for every energy. Requiring the same of the vertex $V = I + w\sigma 1$ gives $\sigma 1(I + w\sigma 1)\sigma 1 = I + w^{*}\sigma 1$, hence $w = w^{*}$, hence the fold amplitude is real.

The contrast with the spatial case is exact and is the technical heart of the construction. In Feynman's spatial checkerboard the two components are directions of motion in space, the positive form at a cell is the entire probability content of that cell, and demanding $V^{\dagger}V = I$ gives $w + w^{*} = 0$, so the fold amplitude is imaginary, the vertex lies in SU(2), and the group is compact. In the temporal construction no such probability is available in advance, reversibility is what is available, and it delivers the opposite phase. The vertex lies in U(1,1), and it satisfies $V^{\dagger}\sigma 3 V = \sigma 3$ identically while $V^{\dagger}V \neq I$.

Everything else is a consequence. Expanding the step $T = VD$ to first order with $w = \Delta\varepsilon/\hbar$ gives the generator

$$H = E\,\sigma 3 + i\,\Delta\,\sigma 1,$$

the algebra su(1,1) rather than su(2), the conserved indefinite bilinear $Q = |\psi+|^2 - |\psi-|^2$, the invariant $E^2 - \Delta^2 = -\det H$, and the spectral timescale

$$\tau RC = \hbar / \sqrt{(E^2 - \Delta^2)} \quad (E > \Delta), \quad \tau amp = \hbar / \sqrt{(\Delta^2 - E^2)} \quad (E < \Delta).$$

The two theories therefore stand at the two ends of one continuous rotation of the phase of w. At $\arg w = \pi/2$ the amplitude oscillates through exact zeros and the theory is Dirac's. At $\arg w = 0$ successive folds add rather than cancel and the theory is the time-symmetric one. In second-quantised language the distinction is

between a beam splitter and a parametric amplifier: a Hermitian coupling transfers quanta and conserves their number, an anti-Hermitian coupling creates and destroys pairs and conserves the difference, and the transformation is the Bogoliubov one.

The same generator by an independent route, and the origin of the metric

Eliminating one component returns, with no approximation, a second-order equation of Klein-Gordon type whose mass term is the invariant $E^2 - \Delta^2$. We then reverse the procedure and ask what first-order generator squares to that invariant. Writing $H = E\Gamma 1 + \Delta\Gamma 2$ and requiring $H^2 = (E^2 - \Delta^2)I$ for all E and Δ gives the signature conditions

$$\Gamma 1^2 = +I, \;\; \Gamma 2^2 = -I, \;\; \{\Gamma 1, \Gamma 2\} = 0,$$

whose smallest solution is $\Gamma 1 = \sigma 3$ and $\Gamma 2 = i\sigma 1$. The generator obtained by counting folded paths is recovered by demanding that a square root be linear, and the agreement of the two derivations is the check that matters.

The metric operator arrives with those conditions rather than being imposed. Since $\Gamma 1$ squares to the identity it may be taken Hermitian, while $\Gamma 2$ squares to minus the identity and must be anti-Hermitian, so conjugation by $\Gamma 1$ gives $\Gamma 1 H \Gamma 1^{-1} = H\dagger$. The metric is $\eta = \Gamma 1 = \sigma 3$, and the conserved form is indefinite precisely because $\Gamma 2$ had to be anti-Hermitian. Pseudo-Hermiticity is a consequence of the signature and not a further postulate. The difference from Dirac is one sign: Dirac requires both generators to square to plus the identity, the temporal linearization requires one of each.

The interpretation of Q follows Feshbach and Villars and Pauli and Weisskopf, and we adopt it without softening. Q is not a probability. In two-component form the theory carries both an indefinite bilinear and a positive one on the same space, only the first is conserved by the free dynamics, and second quantization returns a positive-definite norm on Fock space. Q is a branch charge, the net imbalance between the two senses of traversal along a worldline, in exact analogy with the imbalance between particles and antiparticles.

What noncompactness does and does not buy

Amplification is not simply what a noncompact group delivers, and we are explicit about the limits of the claim. For any diagonalizable pseudo-Hermitian generator with real spectrum there exists a positive-definite metric in which the evolution is unitary, by the quasi-Hermitian construction of Scholtz, Geyer and Hahne and of Mostafazadeh. Above threshold, $E > \Delta$, that metric exists and is constructed explicitly, so growth of the positive form there is a statement about a choice of inner product and not an invariant of the dynamics. No violation of unitarity is claimed anywhere.

Invariant growth exists only in the broken phase $E < \Delta$ and at its boundary, where the spectrum is imaginary, the pseudo-Hermiticity is spontaneously broken in the sense familiar from PT-symmetric quantum mechanics, and no positive-definite metric exists. Any observable consequence of fold dynamics is therefore traceable to the neighbourhood of the exceptional point $E = \Delta$ or below it.

That confinement carries a division of labour we regard as the principal structural result for foundations. Identifying Δ with mc^2 makes $E \geq \Delta$ hold for every momentum, with equality only at rest, so free evolution never enters the amplifying sector. Reaching $\Delta > E$ requires something to drive the mode across the band edge, and in the measurement context that something is the coupling to the apparatus. The kinematics, meaning the two branches, the indefinite charge, the noncompact group and the interference time, follows

from closed reversible dynamics with no measurement anywhere in the argument. The symmetry breaking is dynamical and must be caused. Tunnelling is the ordinary and already observed face of the same structure, since below a mass gap the solutions of a Klein-Gordon equation are evanescent while the conserved current stays fixed and the density does not.

The elementary interval, and why no deformation of the Lorentz group is required

Equating the gravitational self-energy, the rest energy and the confinement energy of a particle localised in a region of size ℓ fixes that size uniquely at the Planck length, with no field equations entering the argument. The quantity discretized, however, is not a coordinate. It is the arc length of the worldline in the sense Stueckelberg gave it, which is a scalar.

That choice settles the question that a discrete construction ordinarily raises. A lower bound on proper time is inherited by every observer in the strengthened form $\Delta t = \gamma\tau \geq t_p$, so a boost can only reinforce it. A lower bound on a rest length is assigned $\ell 0/\gamma$ by a moving observer and fails for every γ above $\ell 0/\ell p$, so restoring it demands a cutoff on γ, a deformed boost, doubly special relativity, and a noncommuting coordinate algebra as a consequence rather than as a free choice. Discreteness of proper time costs nothing and discreteness of length costs the Lorentz group. The Snyder algebra is the price of a discrete position spectrum with an undeformed Lorentz algebra, and no such price is incurred by discretizing a scalar.

The comparison with κ-Poincaré is carried out explicitly, since both Casimirs are in hand. The κ-Poincaré correction is first order in the inverse Planck energy, independent of mass, and survives at zero mass, so it deforms the photon dispersion. The lattice correction is second order, proportional to Δ^2, saturating above threshold, and identically zero for a massless mode. The two disagree in order, in mass dependence and in energy dependence, so no bicrossproduct structure and no coordinate algebra is inherited, and the construction lies outside the reach of the gamma ray burst timing bounds that constrain first-order deformations.

Two further consequences of setting the cell to the Planck cell deserve mention. The fold amplitude is capped, $|w| < 1$, giving $\Delta < E_p$ from the counting rather than by assumption, and with $\Delta = mc^2$ the amplitude per tick is the ratio of the Planck time to the Compton time, so the reduced Compton length is the mean spacing between folds along a worldline rather than a size. The energy enters only as an angle, so the band has two edges, and the interference time acquires an exact floor of t_p/π attained at the upper edge. A collapse criterion phrased as the driving of τ_{RC} to its minimum therefore selects the upper band edge and identifies measurement with the exhaustion of the available temporal resolution.

Numerical verification

Every algebraic claim is checked against exact lattice transfer matrices. The trichotomy of SU(1,1) reproduces the continuum threshold at $E = \Delta$ exactly. The vertex with real fold amplitude preserves σ_3 to 2×10^{-16} and is not unitary, the vertex with imaginary amplitude is unitary to 10^{-16} and does not preserve σ_3, and the Θ-covariance holds identically for the first and fails for the second. In the hyperbolic regime the positive form grows by a factor of 1.87×10^7 while Q is held to 1.6×10^{-8}, which is the arithmetic floor of double precision for a difference of that size. Exponential growth of one quadratic form alongside constancy of another to the limit of the arithmetic cannot happen by accident.

Relation to existing time-symmetric work

Aharonov, Bergmann and Lebowitz established that the formalism admits a symmetric description of preparation and selection, and the two-state vector formalism with its weak values developed that description into a calculus with measurable consequences. Cramer's transactional account treats emission and absorption as one event. Wharton and Argaman survey the locally mediated reformulations. Huw Price took the general case furthest, arguing that microscopic temporal asymmetry is an assumption and not a datum, that it is inherited from the macroscopic thermodynamic asymmetry by an inference that has not been justified, and that Bell's theorem constrains locality only if that inference is granted.

The construction was reached independently and by another route, from the absence of a consistent microscopic definition of the direction of time, and it arrives in territory Price had already mapped. What differs is what is taken as given. No post-selected second state vector appears and no weak values are computed, the symmetry at issue is a symmetry of local vertex weights on a lattice rather than a boundary condition imposed at two times, and the axioms are outputs rather than inputs.

Open problems

Four items are open, and we do not disguise them. The identification $\Delta = mc^2$ is guided by the analogy with the spatial checkerboard and is not a theorem; the lattice supplies the ceiling and the invariant and does not supply the value of Δ for a given system. Proper time degenerates on null curves, so the scalar carrying the discreteness for massive systems has no massless counterpart, and the covariant completion in which four-volume replaces arc length has not been carried out. The absence of signalling is verified case by case and is not derived, which is the fine-tuning objection in its technical form; a conserved branch charge held at every vertex carries material for a reply, and the reply remains to be constructed. Deciding empirically between the two theories is the fourth, and the accessible arenas are the Rydberg-scaled hydrogen deviation $\Delta E_n/|E_n| = (\Delta_0^2/2\mathrm{Ry}^2)n^4$, whose n^4 scaling is the most testable prediction of the scheme, and SU(1,1) interferometry of the Yurke type, where the boost obtained at the vertex is the operation the apparatus performs.